MILITARY SPECIFICATION

VACUUM PUMPS, POWER DRIVEN, CENTRIFUGAL PUMP

PRIMING SERVICE, NAVAL SHIPBOARD

1. SCOPE

1.1 This specification covers the design and construction of vacuum pumps for the purpose of priming centrifugal pumps by removing air and associated vapors from pump casings and suction piping.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular, (General Purpose).
QQ-C-390 - Copper Alloy Castings (Including Cast Ear).
QQ-N-281 - Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
GGG-P-781 - Puller, Mechanical Puller Attachment, Mechanical, and Puller Set, Mechanical.

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MIL-B-857 - Bolts, Nuts, and Studs.
MIL-S-901 - Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.
Mm-D-l 000 - Drawings, Engineering and Associated Lists
MIL-D-1000/2 - Drawings, Engineering and Associated Lists.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval shipboard .
MIL-P-15024 - Plates, Tags and Bands for Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements For.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment Naval Shipboard Use).
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-E for Metals).
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.
MIL-M-17059 - Motors, 60 Cycle, Alternating Current, Fractional H.P. (Shipboard Use).
MIL-M-17060 - Motors, 60 Cycle, Alternating Current, Integral Horsepower (Shipboard Use).
MIL-M-17413 - Motors, Direct-Current, Integral H.P. (Shipboard Use).
MIL-P-17545 - Primer Coating, Alkyd-Red Lead Type, Formula No. 116 and Formula No. 116D.
MIL-M-17556 - Motor, Direct-Current, Fractional H.P. (Shipboard Use).
MIL-M-38761 - Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System General Requirements for, Preparation of.
MIL-M-38761/2 - Microfilming and Photographing of Engineering/Technical Data and Related Documents: PCAM Card Preparation, Engineering Data Micro-Reproduction System: Microfilm Aperture and Tabulating Cards for Naval Ship Systems.
MIL-I-45208 - Inspection System Requirements.

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STANDARDS

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MIL-STD-278 - Fabrication, Welding, and Inspection of Machinery, Piping, and Pressure Vessels for Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service,
MIL-STD-777 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Surface ships .

DRAWING

MILITARY

B-153 - Packings and Gaskets, Standard Application of.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein, Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office Washington, D. C. 20402).

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules,

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606),

3. REQUIREMENTS

3.1 Qualification. Power driven vacuum pumps furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.2 and 6.3).

3.2 Materials. Materials for the pump shall be in accordance with table I.

Table I - List of materials.

Name of part	Material	Specification
Casing	Gun metal	QQ-C-390, alloy D5
Rotor	Ni-Cu alloy, comp. A or gun metal	QQ-N-288 or QQ-C-390
Port plate or cylinder	Bearing bronze	QQ-C-390, alloy E7
Gland	Gun metal	QQ-C-390
Shaft	Ni-Cu-Al alloy or Ni-Cu alloy	QQ-N-281
Shaft sleeve	Ni-Cu-Al alloy	QQ-N-281
Mounting brackets	Gun metal	QQ-C-390, alloy D5
Seal water tank	Galvanized steel	- - - -
Priming valve casing	Gun metal	QQ-C-390
Priming valve float	Ni-Cu alloy, comp. A or rubber	QQ-N-288 or QQ-C-390

3.2.1 Specification QQ-C-390 covers a number of alloys. Alloys other than those shown in table I may be used subject to prior approval by the Naval Ship Engineering Center (NAVSEC).

3.2.2 The use of materials differing in properties from those specified for a given application will be considered when the supplier shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the command or agency concerned.

3.2.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the pump. The use or substitution of manufacturers material shall be subject to approval by the command or agency concerned and shall

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be clearly indicated in a list submitted with the master drawings.

3.3 General design.

3.3.1 The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. Each pump shall be designed to have a service life of 20 years of which 10 percent shall be actual operation: There shall be no limit on the number of starts during the life of the pumps. Parts subject to wear and deterioration (with the exception of packing and seals) will be replaced at intervals not shorter than three years. The requirements specified above shall not be construed or interpreted as a warranty requirement nor otherwise affect the manufacturer's warranty. The design and construction of all pumps shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion,
(d) Economy.
(e) satisfactory operation when inclined as follows:
(1) For surface ships:
a. Up to 15 degrees to either side (permanently inclined).
b. With the ship rolling up to 45 degrees from the vertical to either side.
c. With the ship pitching 10 degrees up and down from the normal horizontal plane.

(2) For submarines:

	Submerged	Snorkel	Surface
Trim	±30 degrees	±7 degrees	±7 degrees
Pitching	±7-1/2 degrees	±10 degrees	±10 degrees
Time of complete pitch cycle	120 seconds	6 seconds	6 seconds
List	15 degrees	15 degrees	15 degrees
Rolling	±30 degrees	±30 degrees	±60 degrees below 10 knots and ±30 degrees above 10 knots
Time of complete roll cycle	14 seconds	14 seconds	12 seconds
Snap roll	55 degrees		
Time of snap roll	20 seconds		

Note: Degrees measured from the vertical to either side for list and roll, and from the normal horizontal plane up or down for trim and pitch.

3.3.2 The priming pump shall be of the centrifugal displacement, wet vacuum type and shall be provided with all necessary fittings and accessories including vacuum relief valve, check valve and similar items required for the satisfactory priming of one or more centrifugal pumps as required by the contract or order.

3.3.2.1 Each priming pump shall be furnished with an automatic valve to prevent flow of liquid from the centrifugal pump or piping system to the primer. The automatic priming valve shall be of a rugged simple design without levers, clevis pins or parts which would have a tendency to vibrate or wear.

3.3.2.2 The design of the primer shall be such as to keep metal contact to an absolute minimum in order to reduce wear.

3.3.2.3 Consideration will be given to priming pumps of designs other than the centrifugal displacement Unit vacuum type providing all other features required herein are retained; the performance is equal to or better than that of the centrifugal displacement wet vacuum type; and the weight, space, and power requirements do not exceed those of the centrifugal displacement wet vacuum type,

3.3.3 Mounting.

3.3.3.1 Pump units shall be horizontally mounted.

3.3.3.2 When specified (see 6.1), the pump unit shall be mounted on a bedplate which also serves as reservoir for the seal water.

3.3.3.3 Special attention shall be given to the design of closed-coupled pump to insure that leakage from the pump stuffing box glands cannot enter motor frames or bearing brackets.

3.3.4 Bearings.

3.3.4.1 Radial and thrust bearings shall be of the ball type.

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3.3.4.2 The use of ball bearings shall be subject to the speed limitations of FF-B-171 except where prior approval of the command or agency concerned has been granted.

3.3.4.3 Installation.

3.3.4.3.1 Bearings shall be installed in housings separate from and independent of the stuffing boxes and with adequate protection from gland leakage. Suitable means shall be provided-to prevent escape of lubricant around the shaft. All baffles, wipers, and related parts shall be readily renewable.

3.3.4.4 Lubrication of the ball bearings shall be in accordance with FF-B-171.

3.3.5 Motors.

3.3.5.1 All motors for driving pumps shall be of the types and characteristics specified (see 6.1), and shall conform to MIL-M-17413 or MIL-M-17556 for direct current (d.c.) motors, or MIL-M-17059 or MIL-M-17060 for alternating current (a.c.) motors.

3.3.5.2 The horsepower rating of each motor shall be not less than the maximum brake horsepower of the driven pump under any condition from shut-off to free delivery. The actual motor rating shall be in accordance with Navy standard motor ratings, If the maximum brake horsepower of the pump is less than 2, and if there is a possibility of an unpredictably high frictional load due to improper adjustment of the gland or to some other cause, then the maximum brake horsepower shall be multiplied by a safety factor 1.5 - $\frac{\text{(max. BHP)}}{4}$ before selecting the next larger Navy standard rating,

3.3.6 Controllers. All motor controllers shall be of the characteristics specified (see 6.1) and shall conform to MIL-C-2212 for both a.c. and d.c. controllers,

3.3.7 Shockproofness.

3.3.7.1 The design of all complete pumping units shall be such that they are capable of passing the grade A, hull mounted, class I, medium weight, type A-shook requirements specified in MIL-S-901 (see 4.4.7).

3.3.7.2 Shockmounts shall not be used unless command or agency approval of the mount and its proposed application is given.

3.3.7.3 The determination of stresses resulting from the shock tests is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in parts due to shock shall be the yield strength (0.2 percent offset of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up shall not be added to the shock stresses. It is of paramount importance that the maximum accelerations possible under the shock tests be utilized in the design of holding down bolts, supporting feet or lugs, and main structural members of the equipment, and that no failure or permanent deformation occur as a result of shocks. Parts of an assembly mounted on a common rigid base need not be designed for higher accelerations than the values for which the whole assembly is designed unless the designer has reason to believe that higher accelerations will be experienced by the parts.

3.3.7.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance. Hole diameters shall be no more than 1/32 inch larger than the bolts for sizes UP to and including 3/4 inch, and no more than l/16 inch larger than bolts of 3/4 inch size.

3.3.8 Painting.

3.3.8.1 For pumps operating at a temperature of 300°F. or lees, all external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with one coat of pretreatment, in accordance with MIL-P-15328, and one coat of zinc chromate primer in accordance with TT-P-645, or one coat of red lead primer in accordance with MIL-P-17545, followed by a finish coat of light gray equipment enamel in accordance with MIL-E-15090.

3.3.8.2 Painting of external surfaces of nonferrous parts of pumps will not be required,

3.3.9 Threaded parts such as bolts, studs and nuts shall conform to MIL-B-857 and Handbook H28.

3.3.10 Welding shall be in accordance with MIL-STD-278 except for the welding of the seal tank.

3.4 Detail requirements.

3.4.1 Casings.

3.4.1.1 The easings of all pumps shall be designed giving maximum consideration for accessibility and ease of replacement of packing, mechanical scale, bearings, and wearable parts such as rotors.

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3.4.1.2 Sufficient clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools,

3.4.1.3 Fitted bolts or dowel pins shall be provided to insure maintenance of alignment of easing parts ‘in re-assembly.

3.4.1,4 Casings shall be provided with drain connections.

3.4.2 Suction and discharge connections.

3.4.2.1 Suction and discharge connections shall be flanged for mating with flanges in accordance with MIL-STD-438 or MIL-STD-777 as applicable, and as specified (see 6.1), except tapered pipe threads are acceptable for use in grease fittings,

3.4.2.2 The thichness of the flanges shall be not less than that specified in the applicable specification or drawing.

3.4.3 Rotors and shafts.

3.4.3.1 Rotor surfaces shall be smooth finished,

3.4.3.2 Rotors shall be keyed on the shaft and securely held against lateral movement by looked nuts or other means approved by the command or agency concerned.

3.4.3.3 Shaft threading shall be counter to the direction of rotation except where both right hand and left hand units of the same design are involved, in which case, in order to reduce the number of shafts required for repair parts, the threading shall be right hand for all unite and the threaded parts shall be looked in place.

3.4,3.4 Shafts shall be fitted with shaft sleeves in way of stuffing boxes unless otherwise approved by the command or agency concerned,

3.4.3.5 The shafts of all pumps, except those driven by a.c. motors, shall be provided with means to permit the ready and direct use of portable tachometers.

3.4.4 Stuffing boxes and packing.

3.4.4.1 Stuffing boxes shall be of adequate depth and design to reduce leakage to a minimum under all operating conditions. Mechanical seals are acceptable.

3.4.4.2 Packing shall be in accordance with Drawing B-153. All packing shall be of a brand approved by the command or agency concerned.

3.4.4.3 The design shall be such as to insure that leakage from the glands cannot reach the bearings or be thrown over bearing housings or driving units. This shall preferably be accomplished by using glands cored out internally and the shafts provided with flingers. Ample drip pockets, spray shields and drains shall be provided.

3.4.4.4 Glands shall be split in halves with the halves secured together in a reamer to form solid rings for setting up.

3.4.4.5 All stuffing box glands shall be set up by units threaded on gland swing bolts, or on studs secured in the casings.

3.4.4.6 Mechanical seals when used shall be positioned on the shaft by means of stub or step sleeves, Mechanical seals shall not be positioned by use of set screws. The design shall insure that positive liquid pressure is supplied to the seal faces under all conditions of operation and that there is adequate circulation of the liquid to the seal faces to minimize the deposit of foreign matter on the seal parts,

3.4.5 Couplings.

3.4.5.1 Flexible couplings, when required, shall be of all metal construction.

3.4.5.2 All flexible coupling hubs shall be keyed to the shafts and secured by locknuts. For shafts 1 inch in diameter and larger, the hubs shall be fitted on a taper. Horizontally flexibly coupled units with bedplates shall have coupling guards provided. Guards shall be designed to permit ready access to the coupling for lubrication and Inspection.

3.4.6 Priming valve.

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3.4,6,1 Priming valves shall be of the float type and shall permit the flow of air while the pump is being primed, shall close when the water level reaches the valve to prevent water from entering the priming pump, and shall open whenever the centrifugal pump loses its prime.

3.4.6.2 Inlet and outlet connections shall be flanged in accordance with MIL-STD-438 or MIL-STD-777, as applicable.

3.4.7 Seal water tanks.

3.4.7.1 The seal water tank shall be of welded construction with suitable reinforcements to accommodate the weight of the priming pump unit.

3.4.7.2 The tank shall be fitted with a sight glass, filling connection, drain connection, vent and over-flow connection, and hand hole.

3.5 Identification plates.

3.5.1 Identification plates shall be furnished for each pump and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic or aluminum are not acceptable.

3.5.1.1 A separate identification plate shall be furnished for the priming valve,

3.5.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.5.3 Pump identification plates shall contain data as follows:

(a) Manufacturers name.
(b) Manufacturers model or type and size.
(c) Service application (that is, vacuum priming pump).
(d) Manufacturers serial number.
(e) Salient design characteristics:
    (1) Capacity in cubic feet per minute.
    (2) Vacuum in inches of mercury.
    (3) Speed in revolutions per minute.
    (4) Brake horsepower.
(f) Contract number.
(g) Federal stock number.
(h) Section for inspectors stamp.

3.5.4 Priming valve identification plate shall contain data as follows:

(a) Manufacturers name.
(b) Manufacturers model or size.
(c) Manufacturers serial number.
(d) Service application (that is, vacuum priming valve),
(e) Contract number.
(f) Federal stock number.
(g) Section for inspectors stamp.

3.5.5 Each driving unit and accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.6 Drawings.

3.6.1 Drawings shall be in accordance with categories E, F, G and H, form 2 of MIL-D-1000 and types II and III of MIL-D-1000/2.

3.6.2 Outline drawings shall be furnished covering complete pump units including pump, driver, bedplate and attached accessories. Separate outline drawings shall be furnished for each contract or order and for each different pump unit on a contract or order.

3.6.2.1 Performance curves on preliminary outline drawings as originally submitted to the command or agency concerned may be calculated curves and shall be so indicated. The complete performance curves on the validated outline drawings shall be based on the performance tests and shall be so indicated. When a number of duplicate pumps are furnished under one contract, all pumps shall be given a performance test, and provided all meet or exceed design conditions, only the performance curve for one unit shall be shown on the drawings.

3.6.3 Pump master drawings shall include sectional assembly, complete list of material and details of all parts. The section assembly and list of material with references to detail drawings should be on a single

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sheet; and details shall be on one or more additional sheets. Several details should be shown on each pump master drawing. Details of single pump parts on size A sheet, resulting in a multiplicity of drawing sheets, should be avoided, The sectional assembly drawing shall show all assembly clearances,

3.6.4 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.6.5 Drawings for drivers and associated equipment shall be in accordance with specifications for those equipments.

3.6.6 After reconciliation of any comments on drawings, final pump and motor drawings shall be submitted. Distribution of final drawings shall include special requirements of the contract or order and the following:

(a) One set of aperture cards and one set of tabulating cards in accordance with MIL-M-38761 and MIL-M-38761/2 to Navy Publications and Printing Service Office, Building 157, Washington Navy Yard, Washington, D. C. 20390.

(b) One set of aperture cards in accordance with MIL-M-38761/2 and one set of folded prints or reduced size prints to the Naval Ship Engineering Center, Center Building, Prince George's Center, Hyattsville, Maryland 20782.

3.7 Technical manuals. Technical manuals shall be in accordance with type I of MIL-M-15071 unless otherwise specified in the contract or order (see 6.1). A separate technical manual shall be furnished for each different pump unit except in special oases when manuals covering more than one pump may be approved. Preliminary technical manuals submitted for approval shall include all proposed sections complete.

3.8 Repair parts and tools.

3.8.1 Onboard repair parts of the types and quantities shown in table II, plus any additional parts considered necessary to properly maintain the pump shall be furnished as required in the contract or order (see 6.1).

Table II - Onboard repair parts.

Name of part	Quantity per set
Rotor	1
Rotor key	1
Shaft sleeve	1
Packing	1 set
Water flingers and set screws	1
Casing (excluding mounting bracket)	1
Gaskets	1 set
Locknuts and washer for rotors	1
Coupling wearing parts	1 set
Priming valve float	1
Priming valve bushing	1
Priming valve seat	1
Priming valve gaskets	1 set
Relief valve	1

(When flexible coupled or belt driven units are furnished, one complete rotating assembly should be furnished in lieu of the duplicate parts listed herein. The coupling or sheave-should be included.)

3.8.1.1 Repair parts for the drivers and associated equipments shall be furnished in accordance with the applicable equipment specification. Motor rotors or armatures shall be furnished for close-coupled units.

3.8.1.2 A complete set of special tools required for the maintenance and repair of the pump units shall be included in each set of onboard repair parts. Special tools are defined as those tools not listed in the Federal Supply Catalog (copies of this catalog may be consulted in the office of the Defense Contract Administration Service (DCAS)).

3.8.1 .2.1 Tools which are available in the Federal Supply Catalog, such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

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3.8.1 .2.2 Each box containing onboard repair parts-shall contain a list entitled "List of onboard repair parts and tools". The list shall be in a format suitable for use on outline drawings and in the technical manual s , It shall be not less than nominal 8-1/2 by 11 inches in size.

3.8.1 .2.2.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number_____
Number of ships_____
Application_____
Contract number_____
Manufacturer_____
Quantities are for_____ units per ship

The columns shall include:

Piece' number_____
Name of part_____
Quantity_____
Drawing number_____
Federal stock number_____
Additional columns may be used as applicable.

3.8.1.2.2.2 The list shall be so treated as to be resistant to oil, water and fading.

3.8.2 Stock. The selection, stock numbering and quantities or repair parts for stock shall be determined and processed in accordance with MIL-P-15137 (see 6,2).

3.8.2.1 Unless otherwise specified in the contract or order, the cognizant inventory control point for repair parts for stock is Ships Parts Control Center, Mechanisburg, Pennsylvania.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirement.

4.1.1 Inspection system. stem. The supplier shall provide and maintain an inspection system acceptable to the Government for supplies and services covered by the specification. The inspection system shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

4.2 Qualification tests Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. The characteristics of the units for qualification tests shall be as specified by NAVSEC. The qualification tests shall consist of the tests specified in 4.2.1 through 4.4.7.6,

4.2.1 The qualification tests shall adequately demonstrate the ability of the pump to handle the specified capacity at the specified vacuum. Sufficient data shall be taken during the tests to prepare pump characteristic curves as required herein. Test data shall be complete over the entire range of capacities from shut-off to as near free delivery as possible. The following test curves shall be prepared:

(a) Vacuum versus air capacity.
(b) Vacuum versus brake horsepower.

4.2.2 Endurance test. The pump shall be operated for 24 hours continuously at a vacuum as close to design as possible and for 12 hours against a blank suction. Failure of any part of the pump during the endurance test shall be considered as failure of the pump to qualify. The pump shall complete the endurance test without excessive noise, vibration or wear of parts.

4.2.3 In addition to the tests at the laboratory, Pumps may be required to pass the following shipboard service tests:

(a) Two or more pumps of the general design and general operating characteristics shall have given Satisfactory service onboard a Naval ship for a period of at least two years and shall still be in service on such ship at the time qualification is granted.

1/ Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD- (see 6.3 and 6.3.1).

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4.3 Shop tests.

4.3.1 Each component part listed below for each vacuum pump unit shall be tested hydrostatically to a pressure of at least 150 percent of the maximum pressure to which it can be subjected in operation, but not less than the pressures listed as follows:

Pump casings	50 p.s.i.g.
Priming valve body	30 p.s.i.g.
Priming valve float	25 p.s.i.g.
Seal tank	15 p.s.i.g.

4.3.2 Each pump, except those driven by a.c. motors, shall be tested in the shop of the manufacturer or supplier at 25 percent above the designed maximum operating speed by a continuous nonstop run of at least 30 minutes. This test shall be conducted in the presence of the Government representative, who shall check operation and smoothness of running. In the case of pump driven by a.c. motors each unit shall be tested for smoothness of operation at full speed.

4.4 Performance tests.

4.4.1 It shall be the responsibility of the prime contractor to insure that tests required on assembled units are made.

4.4.2 Driving units shall be tested as required by the applicable specification,

4.4.3 Performance tests shall adequately demonstrate the ability of the pump to handle its rated capacity of air at the specified vacuum.

4.4.4 sufficient data shall be taken during the tests to prepare pump characteristic curves as specified in 4.4.5. Test data and curves shall be complete over the entire range of opacities from shutoff (highest vacuum obtainable) to as near free delivery (atmospheric pressure) as possible.

4.4.5 The following test curves shall be supplied for consist speeds of approximately 90, 100 and 110 percent of designed speed:

(a) Vacuum versus air capacity.
(b) Vacuum versus capacity in free air.
(c) Vacuum versus brake horsepower.

Where pumps are driven by a.c. motors, the curves shall be obtained for the rated speed condition of the motor in lieu of those specified above.

4.4.6 Complete test reports, including test data on both pumps and driving units, shall be assembled and distributed as follows:

(a) Three to NAVSEC (to be forwarded via the cognizant Government representative).
(b) One to Defense Contract Administration Services District.
(c) One to Defense Contract Administration Services Office.
(d) Two to each Supervisor of Shipbuilding (one for supervisor's files, the other for the shipbuilder).
(e) One to the Commander of each shipyard concerned, for Naval shipyard built ships, when specified in the contract or order.

4.4.7 Shock tests.

4.4.7.1 One complete pumping unit of each design shall be subjected to the high impact shock tests specified in MIL-S-901 (see 3.3.7).

4.4.7.2 Equipment previously accepted will not be resubmitted for shock test except when evidence of low shock resistance develops in the units installed. The command or agency concerned will initiate action with the manufacturer for correction of deficiencies, and reserves the right to require shock tests at any time.

4.4.7.3 All PUMP units shall be shock tested with drivers unless otherwise approved by the command or agency concerned. Flexible coupled pumps shock tested with one driver will not be required to be shock tested again when supplied with a different driver. Prime movers are subject to shock tests in accordance with the applicable equipment specification.

4.4.7.4 The pump unit shall be considered to have failed to pass the shook tests in the event of any of the following:

(a) Breakage of any parts, including mounting bolts.

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(b) Appreciable distortion or dislocation of any part, such as shaft, mounting feet, bearings.
(c) A mechanical unbalance of more than two times the amplitude of unbalance measured prior to tests at rated speed, or more than that permitted by the specifications for the driver.

4.4.7.5 Pump units which have been subjected to the high-impact shock test and have failed to conform to the requirements will not be acceptable.

4.4.7.6 Pump unite which have been subjected to the high-impact shock test and have successfully passed this test shall be considered acceptable for such service as the command or agency concerned may authorize. Such equipment will be returned to the supplier for reconditioning as follows:

(a) Minor deformations of mounting flanges shall be corrected.
(b) Minor deformations affecting alignment shall be corrected.
(c) All bearings shall be replaced,
(d) Each Part snail be carefully examined by the supplier and any part whihc he considers substandard shall be replaced,

Minor deformations shall be defined as those which do not cause unqualified rejection of the design under high-impact shock test but which are in excess of the design dimensional tolerances specified on the applicable pump drawings.

4.5 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirement of referenced documents listed in section 2, see 6.4).

5.1 Domestic shipment and early equipment installation and for storage of onboard repair Parts.

5.1.1 Pumps.

5.1.1.1 Preservation and packaging Preservation and packaging which may be the supplier's commercial practice, shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.2 Packing, Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation and may conform to the suppliers commercial practice.

5.1,1.3 Marking. Shipment marking information shall be provided on Interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number. supplier's name and destination.

5.1.2 Onboard repair parts. Onboard repair parts shall be preserved and packaged level A, packed level C and marked in accordance with MIL-P-16789.

5.2 Use of polystrene (loose-fill) material.

5.2,1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken onboard ship.-
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bound fiber, fiberboard or transparent flexible cellular material."

5.2.2 For level A packaging and-level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification,
(b) Rated capacity (c.f.me),
(c) Vacuum (in. Hg.),
(d) Mounting required (see 3.3.3.2).
(e) Whether seal tank or valve is required.
(f) Liquid handled by centrifugal pump if other than water,
(g) Whether close coupled or flexible coupled to driver.
(h) Maximum pressure to which priming valve will be subjected.
(i) Electrical characteristics (for motor drive) as required by applicable motor and controller specifications (see 3,3.5,1 and 3.3.6).
(j) Suction and discharge-connections required (see 3.4.2.1).
(k) Drawings required (see 3.6).
(l) Technical manuals required (see 3.7).
(m) Onboard repair parts required (see 3.8).
(n) Quality assurance requirements (see 4,1.1).
(o) When polystyrene (loose-fill) material is approved (see 5.2,1).

6.2 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-P-15137 (see 3.8,2).
(b) MIL-I-45208 (see 4.1.1).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 18683 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturer are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6,3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 CHANGES FROM PROVISIONS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE 0101-807-2000

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Naval Ship Engineering Center
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Prince George's Center
Hyattsville, Maryland 20782

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